

Pizza Sales Analytics & Business Intelligence Report

End-to-End Data Analytics Project using Excel, PostgreSQL & Power BI

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Tools Used: Microsoft Excel, PostgreSQL, SQL, Power BI

Project Type: Data Analytics & Business Intelligence

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1. Executive Summary

This project presents an end-to-end data analytics solution developed to analyze pizza sales data using Microsoft Excel, PostgreSQL, SQL and Power BI. The objective was to transform raw transactional data into meaningful business insights that support data-driven decision-making.

Microsoft Excel was used for initial data preparation, data profiling, validation checks and preliminary exploratory analysis. PostgreSQL and SQL were then used for structured data validation, KPI calculation, sales trend analysis and product performance analysis. Power BI was used to build an interactive dashboard for visualization and business reporting.

The analysis focused on evaluating sales performance, customer purchasing behavior, product performance, revenue contribution and operational trends.

The project demonstrates practical skills in Excel, SQL, PostgreSQL, business analytics, KPI reporting, data visualization and data storytelling, following a real-world business intelligence workflow.

2. Business Problem

A pizza restaurant chain wants to better understand its sales performance and customer purchasing patterns. Although thousands of orders are generated throughout the year, management lacks a centralized reporting system to monitor revenue, identify best-selling products, analyze customer demand, and evaluate business performance.

The objective of this project is to analyze historical sales data and provide actionable insights that can help management optimize operations, improve sales strategies, and support data-driven decision-making.

3. Project Objectives

The primary objectives of this project are:

- Perform initial data profiling and validation using Microsoft Excel.
 - Calculate key business performance indicators.
 - Analyze monthly and daily sales trends.
 - Evaluate pizza category and size performance.
 - Identify top-performing and low-performing products.
 - Discover customer purchasing patterns.
 - Build an interactive Power BI dashboard.
 - Provide business recommendations based on analytical findings.
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4. Dataset Overview

The dataset contains transactional records of pizza orders, including product details, pricing information, order timestamps, quantities, and customer purchasing data.

Each record represents a pizza item sold within an order and includes important business attributes such as pizza category, pizza size, quantity sold, unit price, total revenue, and order date.

The raw dataset was initially reviewed and validated using Microsoft Excel for data profiling, formatting checks, missing-value and duplicate checks, and preliminary exploration. The prepared data was then imported into

PostgreSQL for SQL-based analysis and later connected to Power BI for dashboard development.

5. Data Dictionary

Column	Description
pizza_id	Unique identifier for each pizza sale
order_id	Unique order identifier
pizza_name	Name of the pizza
pizza_category	Pizza category
pizza_size	Pizza size
quantity	Number of pizzas sold
unit_price	Price of one pizza
total_price	Total sales amount
order_date	Date of order
order_time	Time of order
pizza_ingredients	Ingredients used

6. Excel Data Preparation & Validation

Microsoft Excel was used as the initial data preparation and validation layer before performing SQL analysis and developing the Power BI dashboard.

The main Excel activities included:

- Data Profiling and Sanity Checks
- Reviewing Missing Values and Blank Records
- Standardizing Date and Time Formats
- Checking and Removing Duplicate Records where applicable
- Reviewing Data Types and Business Fields
- Preliminary Data Exploration using Pivot Tables
- Validating important fields such as Order ID, Quantity, Unit Price, Total Price and Order Date

Excel provided an initial understanding of the dataset and helped identify potential data-quality issues before loading the data into PostgreSQL.

The overall analytical workflow followed:

Raw Data → Excel Data Preparation & Validation → PostgreSQL / SQL Analysis → Power BI Dashboard → Business Insights

7. Key Performance Indicators (KPIs)

The following KPIs were calculated using SQL and validated through Power BI to evaluate the overall business performance of the pizza sales dataset. Initial data validation and preparation were performed in Microsoft Excel before the SQL analysis.

KPI Summary

KPI	Value
Total Revenue	\$817,621.80
Total Orders	21,334

Average Order Value **\$38.32**

Total Pizzas Sold **49,559**

Average Pizzas per
Order **2.32**

The KPI summary provides a high-level overview of business performance. These metrics serve as the foundation for analyzing revenue generation, customer purchasing behavior, and operational efficiency throughout the project.



Figure 1: Power BI KPI cards displaying Total Revenue, Total Orders, Average Order Value, Total Pizzas Sold, and Average Pizzas per Order.

8. Dashboard Overview

An interactive Power BI dashboard was developed to present key business metrics and sales trends in a visual format. The dashboard uses the prepared and analyzed sales data to provide stakeholders with an interactive view of overall business performance.

The dashboard includes:

- KPI Cards
- Monthly Revenue Trend
- Daily Order Trend
- Sales by Pizza Category

- Sales by Pizza Size
- Product Performance Analysis
- Interactive Filters and Slicers

The dashboard enables stakeholders to explore sales performance dynamically and supports faster business decision-making.

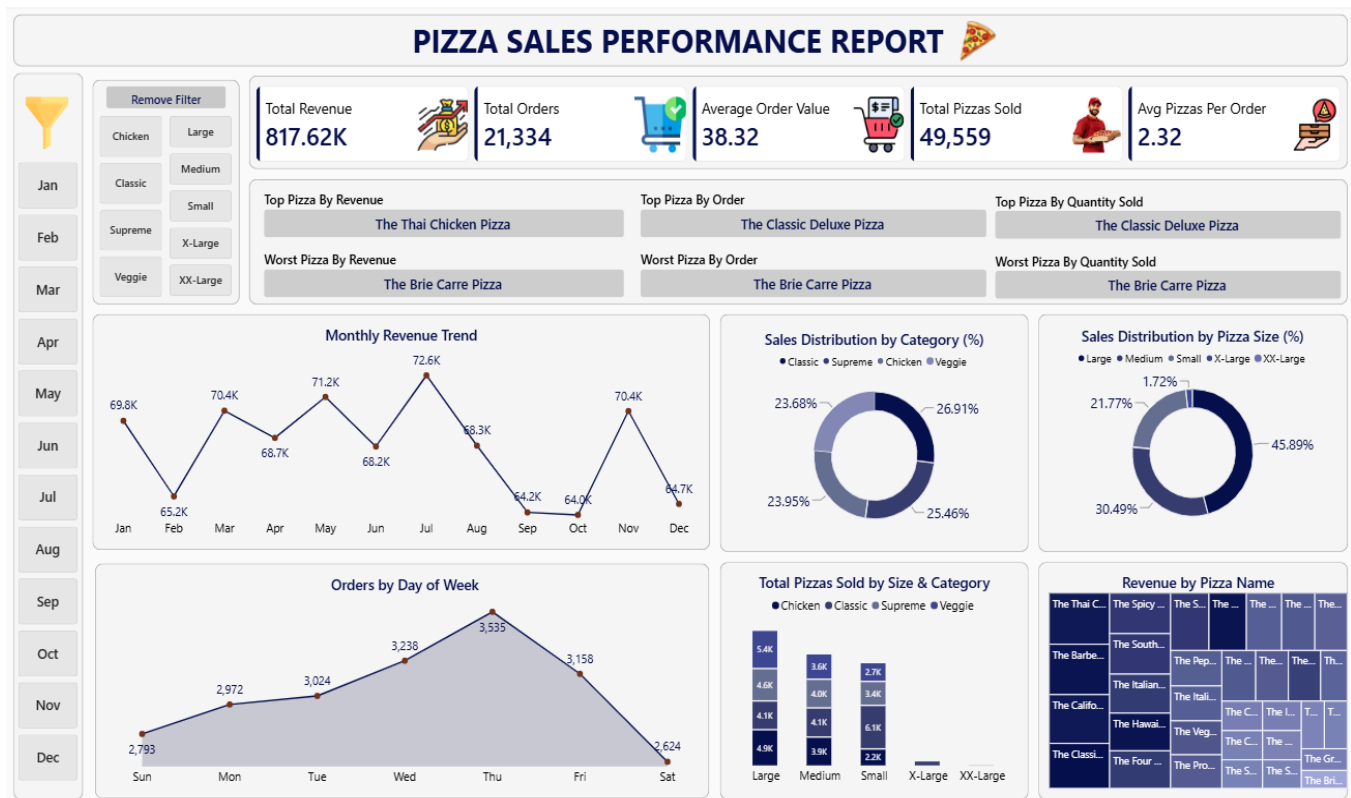


Figure 2: Interactive Pizza Sales Analytics Dashboard developed in Power BI. The dashboard provides an overview of key performance indicators, monthly sales trends, product performance, category distribution, and interactive filtering capabilities.

9. Key Business Insights

Based on SQL analysis and dashboard visualization, several important business insights were identified.

- The business generated strong overall revenue during the analysis period.

- Sales performance varied across different months, indicating seasonal fluctuations.
- Customer ordering activity was higher on specific days and during peak business hours.
- Certain pizza categories consistently contributed a larger share of total revenue.
- Large-sized pizzas represented the highest proportion of sales, indicating customer preference.
- A small number of pizza varieties generated a significant percentage of total revenue, while several products contributed relatively less.
- Peak sales hours can help management optimize staffing and inventory planning.

These findings provide valuable information for improving operational efficiency and increasing profitability.

10. Business Recommendations

Based on the analysis, the following recommendations are proposed.

- Promote top-selling pizzas through combo offers and seasonal campaigns.
- Increase staffing during peak business hours to improve customer service.
- Optimize inventory according to demand for pizza categories and sizes.
- Introduce promotional strategies for low-performing pizzas to increase sales.
- Regularly monitor business KPIs through the Power BI dashboard.
- Use historical sales trends to support demand forecasting and inventory planning.

11. Conclusion

This project demonstrates a complete end-to-end data analytics workflow using Microsoft Excel, PostgreSQL, SQL and Power BI.

Microsoft Excel was used for initial data preparation, profiling and validation, while PostgreSQL and SQL were used for structured data analysis, KPI calculations, sales trend analysis and product performance analysis. Power BI was then used to transform the analytical results into an interactive business intelligence dashboard.

By combining spreadsheet-based data preparation, SQL analysis and interactive visualization, the project successfully transformed raw sales data into meaningful business insights.

The dashboard enables stakeholders to monitor sales performance, identify growth opportunities, evaluate product performance and make informed business decisions.

Overall, this project highlights practical skills in Microsoft Excel, SQL, PostgreSQL, Power BI, KPI reporting, data visualization, business analytics and analytical problem-solving, making it representative of real-world Data Analyst responsibilities.